

Theories of Social Networks

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Abstract and Keywords

Theories of social networks offer abstract perspectives of what social networks are and how they are connected to other features of the social world. This chapter gives an overview of three recent perspectives: (1) Theorists of action (Burt, Coleman, Lin, Hedström) regard social networks as objective structures restricting or enabling individual action. Networks become a resource (social capital) that actors strive to maximize. (2) Authors following pragmatism or symbolic interactionism (Emirbayer, Martin, Crossley) consider social networks as patterns of subjective meaning arising out of the interaction between actors. This approach is linked to field theoretical thinking, considering networks as arising out of the mutual orientation in fields. (3) Relational sociologists (White, Tilly, Mische, Padgett, Fuhse) treat social networks as infused with meaning that is processed in communication/transaction/switchings between actors. Relational sociology has been amended to study networks of symbols and the communicative dynamics of social networks.

Keywords: relational sociology, social capital, social networks, theory, theory of action

Networks and Theory

NETWORK researchers have long lamented a lack of theory of social networks (Granovetter, 1979; Wellman, 1983, p. 179). Social networks were studied empirically, or written about metaphorically, without worrying too much about their substance. Over the last 30 years, a number of researchers have attempted to fill this “theory gap.” This chapter offers an overview of the most important approaches: the theory of action, pragmatist/interactionist approaches, and relational sociology.

Borgatti and Lopez-Kidwell (2011) offer a distinction between “network theory,” on the one hand, and “theory of networks” on the other. They view network theory as about the effects of social networks (e.g., on social mobility), and theory of networks as covering the impacts on networks (e.g., foci of activity). However, Borgatti and Lopez-Kidwell’s

candidates are not really “theory,” but formulations of network mechanisms: systematizing recurrent observable processes involving networks (Wimmer & Lewis, 2010, p. 139ff).

A proper *theoretical* treatment has to take an additional step back and offer answers to the following questions: What are social networks? Why do they constitute important features of the social world? And how do they relate to other features like social categories and other cultural forms, resources and social mobility, formal organizations, and large-scale fields of society? I focus on *social* networks because these are still the most important subject area of network research in the social sciences. Also, social relationships seem to be different from relations between symbolic forms, between actors and material objects, and between events. Therefore, we need different theory to cover these respective networks.

The following short overviews of the most important theories of social networks cover the (1) theory of action with the social capital concept (Burt, Coleman, Lin, Hedström); (2) pragmatist/interactionist approaches (Emirbayer, Crossley, Martin); and (3) relational sociology around Harrison White (Tilly, Mische, Padgett, Fuhse). The concluding section offers a table summarizing these overviews and a few points for further discussion.

(p. 35) Action Theory and Social Capital

The first approach comes from authors adhering to theories of action, following the classic formulation by Max Weber. It comprises Ronald Burt’s, James Coleman’s, and Peter Hedström’s accounts of the *interplay of networks as social structures with individual action*, as well as the notion of *social capital* (Burt, Coleman, Lin). To varying degrees, these authors advocate modeling action on the basis of *rational choice*.

Social Structures and Individual Action

According to theories of action, social phenomena result from individual actions. Decisions for these are taken out of subjective considerations of objective external circumstances. Most often, they conceptualize action as driven by individual calculations of utility: I do whatever promises the best results (according to my interests) in any given situation (Lindenberg, 1990; Hedström, 2005, p. 38ff). Individual actions then lead to the change or reproduction of social structures that constrain or enable future action. Following this general stance of methodological individualism, Ronald Burt (1982) views networks as social structures resulting from individual action, and channeling it in turn (Figure 3.1).

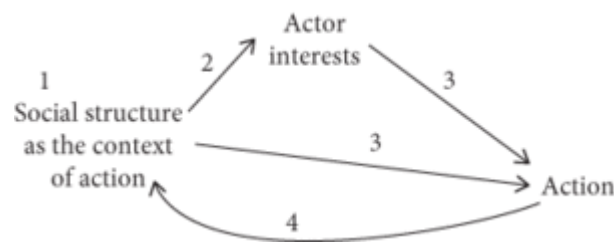


Figure 3.1 Action and social structure according to Ronald Burt (1982, p. 9).

- In any given situation, the network forms part of the *context of action* (1). The position of an actor and his or her ties to others are given and influence his or her behavior.
- The position in the network determines the *actor's interests* (2).
- These interests lead to particular action (3). However, the network not only affects action through interests but also affords actors with different *opportunities* for action, depending on their position and embeddedness (also 3 in Burt's figure). Actors in the center can influence processes in the network, for example, by distributing resources (e.g., money and symbolic esteem). Peripheral actors, in contrast, face less pressure to conform to expectations in the network and enjoy more leeway in their behavior. Networks constrain and enable individual action.
- Finally, individual decisions have *repercussions* for the social structure at hand (4). They can reproduce or change network constellations. Reproduction occurs when action conforms to expectations. Networks change when action deviates from the expected.

(p. 36) James Coleman (1990, p. 8ff) offers a simplified version of Burt's schema with his well-known three-step model of social explanations. Here, the social structure affects only individual decisions, leading to actions that then get aggregated to changes in social structure.

This modeling of networks in action theory has to confront two challenges:

- (1) Networks affect individual actions in two ways: (a) the position in networks enables and constrains action, and (b) networks influence individual preferences or interests (through step 2 in Burt's schema). In theory, models have to incorporate these two mechanisms *simultaneously*, with any one position differing in both opportunities and interests.
- (2) Step 4 in Burt's schema consists in the *aggregation* of actions to a new social structure, or to change in social structure. In statistics, we only have to add numbers to look at rates of individuals acting in isolation from each other (like investing time and opportunity costs in education). In networks, this aggregation requires examining the interplay of a number of actors in different positions, with varying opportuni-

ties and interests for action. Generally, the aggregation of actions in complex social structures constitutes the biggest problem for action theory.

Peter Hedström (2005, p. 76ff) suggests using simulations to examine the aggregation of actions in network constellations. Agent-based modeling supplies individuals in a population with simple decision rules for tie formation, based on their interests and opportunities. However, Hedström's (2005, p. 119ff) empirical example only models distributions of youth unemployment in urban neighborhoods. This instance of a statistical aggregation does not consider the interplay of actors in various network positions.

The SIENA software now allows for agent-based simulations of empirically observed changes in networks (Snijders, van de Bunt, & Steglich, 2010). SIENA is based on action-theoretical assumptions, modeling the formation of ties as the succession of individual decisions. It does not predict (explain) developing network structures. Rather, it looks for individual propensities for the formation of particular ties (e.g., interethnic friendships). While Burt's steps 1, 2, and 3 are modeled with SIENA (to varying extents), step 4 remains out of consideration.

Overall, action theory adopts a naturalistic take on networks and relationships: They are treated as objective structures, without inquiring what they are made of. They result from individual actions, and they structure them in turn. But since multiple social actors participate in their construction, networks and relationships seem to acquire a thing-like quality. Any one individual action has to take them for granted and can only decide to change or to conform to them. In a sense, action theory does not really provide a substantive answer to the question: What are social networks? Rather, it offers a way of modeling their emergence, reproduction, and change. The big advantages of action theory are its intuitive plausibility and its well-established position in the social sciences. Action-theoretical models of networks can build on a host of preliminary work. However, the interplay of actions and networks may be too complex for manageable modeling.

(p. 37) Social Capital

The notion of social capital is also tied to action theory. After the initial formulation by Pierre Bourdieu, a number of action theorists (Coleman, Burt, Lin) adopted the concept. The concept of social capital implies three general assumptions (Lin, 2001, p. 19ff):

- (1) Networks can be seen as a *resource for individuals*. Advantageous network positions and connections afford more opportunities for action, in particular for upward mobility. Social networks are viewed from the perspective of the individual.
- (2) As with economic and human (or cultural) capital, actors possess *more or less* social capital. Therefore, we have to be able to quantify the value of networks for individuals, or at least to distinguish individuals with more social capital from those with less.
- (3) Individuals can *maximize* the value of their social networks. Acting rationally, they should act to accumulate social capital. Individuals enter social relationships

when they promise them net value, and they drop or reject them when expecting more costs than benefits.

These assumptions make for an *economic* perspective on social networks (Somers, 2005): people “invest” in their relations and “capitalize” on them (e.g., when drawing on social connections in their job searches). However, the assumptions remain rather abstract. The authors of the social capital concept have quite different ideas about which ties and networks are particularly valuable, and what we can use them for.

According to *Pierre Bourdieu* (1986), social capital consists in the resources that individuals can mobilize through their social relationships. Social capital can be converted into other forms of capital. Individuals use their social relationships to attain educational degrees or cultural refinement (cultural capital) or for upward professional mobility (economic capital). Since individuals do not really act strategically in Bourdieu’s theory, his concept of social capital only shares the first two assumptions, not the third one. But social capital does not play a systematic role in Bourdieu’s theory. Rather, he disdains network research for a preoccupation with the “surface” of “intersubjective relations” (Bourdieu & Wacquant, 1992, p. 113f).

For *James Coleman* (1988, 1990), the concept refers to the embeddedness of individuals into particular local network structures. Social capital primarily lies in dense networks (with “closure”) that force actors to cooperate with each other. Dense networks solve the “free-rider problem”: recurrent interaction with the same actors allows for sanctioning noncooperation. Therefore, closure in networks leads to the emergence of norms for cooperation and of trust.

Ronald Burt (1992, p. 8ff) also uses the concept for specific network structures. Here, social capital stands for the access of actors to different network clusters (following Mark Granovetter’s [1973] notion of “weak ties”). Actors with “bridges” across “structural holes” in network structure get more information. Hence, they have better chances on markets (in particular, the job market), and they become more creative through combining ideas from disparate networks (Burt, 2004).

Coleman and Burt thus denote contrasting network structures as advantageous. These concepts are brought together in *Nan Lin’s* (2001) theory of social capital. According to Lin, (p. 38) actors possess different forms of social capital: (1) Burt’s bridges across structural holes are important for “instrumental action.” They give access to information from distant areas of social structure. Accordingly, individuals form “weak ties” (without closure) preferably to members of different classes. These primarily help for upward social mobility. (2) Strong ties in dense networks (Coleman’s social capital) assist in “expressive action.” Alters share resources in strong ties (solidarity) and support each other in their worldviews.¹ Therefore, strong ties with closure are primarily found within social classes.

Political scientist *Robert Putnam* (1993) does not treat social capital as an individual resource, but as a property of regional social structures. These differ by their traditionally horizontal or hierarchical network structures. According to Putnam, the horizontal pat-

turning of social relations leads to cooperation and solidarity, and makes for better governance and economic development. Putnam's arguments have become primarily important in the discourse on civil society.

The five approaches sketched here differ substantially in what they term "social capital," in their aims, and in how exactly they are achieved (Table 3.1). Apart from Putnam, all authors reduce networks to *resources of individuals*. The concept focuses on the *effects* of network positions and embeddedness on the individual level. Consequently, it often connects to research on *ego-centric networks*. The conceptualizations by Coleman and Burt (and their combination by Lin) concern the *structure* of personal ties, stressing different aspects as key (dense networks vs. bridges across structural holes). In contrast, these approaches bracket the *composition* of ego-centric networks. Only Bourdieu's theory points to the attributes of the actors connected to, in particular, their resources as important for the value of social relationships.

Theories of Social Networks

Table 3.1 Overview of Concepts of Social Capital

	Bourdieu	Coleman	Burt	Lin	Putnam
What?	Resources that can be mobilized in relationships	Dense social networks (closure)	Weak ties across structural holes	Instrumental and expressive social capital	Horizontal ties (in associations)
What for?	Attainment of economic and cultural capital	Cooperation/solution of free-rider problem	Market opportunities	Social mobility/solidarity and identity	Democratic and economic performance
How?	Mobilization of resources in groups	Sanctions of deviance, trust	Channels of information	Access to information/mutual support	Cooperation and solidarity

Pragmatism and Interactionism

The sociological-psychological school of symbolic interactionism by William Thomas, George Herbert Mead, Herbert Blumer, and Erving Goffman developed out of the (p. 39) philosophical strand of American pragmatism of Charles Sanders Peirce, William James, and John Dewey. Both emphasize the processing of meaning in the mind (pragmatism) and in interaction (interactionism) as the basic layer of social reality. In this vein, Fine and Kleinman (1983) proposed a combination of symbolic interactionism with social network analysis. The most important authors following this lead are Mustafa Emirbayer, Nick Crossley, and John Levi Martin—all three of them also drawing on relational sociology around Harrison White (see Relational Sociology). They are set apart from White's approach by their pragmatist/interactionist foundations in social theory. This entails four important pushes: to a focus on subjective meaning (1), to a processualization of relations and networks (2), to an incorporation of qualitative methods in network research (3), and toward locating networks in social fields (4).

(1) In line with the pragmatist roots, the authors of this approach emphasize the importance of *subjective meaning* for and in social networks. Social relations and networks consist of the subjective meaning that actors hold of them, and this subjective meaning is continuously subject to negotiation and interpretation in interaction (Fine & Kleinman, 1983, p. 101ff). For Crossley (2010a, pp. 28f, 37), the “internalization of the perspective of the other”—the empathy that Mead sees as basic for the process of symbolic interaction—forms part of this subjective construction of relationships. We see ourselves through the eyes of the other and adapt our behavior accordingly. A “common ground” (Merleau-Ponty) or shared “definition of the situation” (Thomas, Goffman) unfolds in the rhythm of interaction. Social relationships build on subjective agreement with regard to how the actors relate to each other.

Emirbayer and Goodwin (1994, p. 1437ff) argue that network analysis and Harrison White's theory of networks fail to take both *agency* and the autonomy of *culture* into account. Culture is not encapsulated in social networks and determined by them. Rather, individuals agentially draw on different cultural ideas to make sense of their situation, and to engage in interactions and relationships with others. With Ann Mische, Emirbayer offers an account of agency that combines habitual iteration, forward thinking, and practical evaluation of alternative courses of action (Emirbayer & Mische, 1998; see also Erikson, 2013, p. 233ff).

In contrast, Martin (2010) builds on recent psychological research to argue that actors are not overly intelligent and creative but follow relatively simple *heuristics* in their behavior toward others. They lead them to connect in particular social network structures—for example, in closed and polarized group structures from heuristics of transitivity in bounded contexts (Martin, 2009), or in ranked prestige hierarchies when people follow others in their choices, as in the sexual field (Martin & George, 2006). These heuristics depend on the relations at hand, on concrete circumstances of the interaction, and on the individual position in relations to others.

(2) All the pragmatist and interactionist approaches focus on subjective meaning as important for social relations and networks. They differ in the modalities of subjective processing of meaning—from creative agency to simple heuristics. In spite of these differences, the authors considered here arrive at similar formulations about the *nature of social networks*. These are conceptualized as “structures of meaning” that develop continuously over the course of interaction (Fine & Kleinman, 1983; Martin, 2009, p. 9ff; Crossley, 2013, p. 124ff). Relationships as the basic building blocks of networks are *processualized*: they are dynamic and fluid (Fine & Kleinman, 1983, p. 99ff), “lived trajectories of iterated interaction” (Crossley, 2010a, p. 28f). Emirbayer (1997, p. 286f) terms the processes at play in social networks “trans-action”: both the identities and the relations between them are constructed (in their meaning) through processes of transaction.²

(3) Some authors following the interactionist tradition have called for the use of *qualitative methods* to study social networks. If social networks are laden with subjective meaning, we have to inquire into this with interpretive methods. Consequently, Crossley (2010b, 2015) combines qualitative tools with the formal techniques of network analysis. Matthew Desmond (2014) lays out “relational ethnography” as a way of examining relational configurations with qualitative interviews and participant observation. In contrast to traditional ethnography, this approach studies “fields rather than places, boundaries rather than bounded groups, processes rather than processed people, and cultural conflict rather than group culture” (Desmond, 2014, p. 548).

(4) The fourth and final push, not directly connected to pragmatist/interactionist roots, locates relations and networks in *social fields*. Martin (2011, p. 268ff) draws primarily on earlier Gestalt psychology for his theory of fields. Social fields are spaces of mutual orientation characterized by uncertainty of the actors at play. As a result, they enter social relations and develop institutionalized rules (DiMaggio, 1986). For Martin (2011, p. 191ff), the behavior of the actors follows simple heuristics (see earlier), or a “social aesthetics” of how to relate to each other. In the case of sexual fields, this results in a prestige hierarchy of actors in terms of their perceived attractiveness (“sexual capital”; Martin & George, 2006).

Building firmly on Pierre Bourdieu, Emirbayer and Desmond (2015) develop the concept of the “racial field.” However, they place less emphasis on networks of intersubjective relations than on Bourdieusian “objective relations” (p. 84ff). These are characterized by the relative distribution of various forms of capital, including the newly discovered “racial capital.” In contrast, social networks feature centrally in Bottero and Crossley’s (2011) account of fields. Building on Bourdieu as well as on Howard Becker’s concept of art worlds, they study the concrete webs of relations among different kinds of actors (artists, managers, venues, customers).

The fourth push may be coincidentally pursued by the most important authors of the pragmatist/interactionist approaches to social networks. The other three pushes—to subjective meaning, processualization, and qualitative-interpretive methods—follow from interactionism and pragmatism. These approaches to social networks seem to have surged over the last years. It remains to be seen to what extent they are adopt-

ed in network research and how they combine with its traditionally formal and quantitative methods.

Relational Sociology

Relational sociology around Harrison White is connected to the pragmatist and interactionist approaches, sometimes interweaving with them (Mische, 2011; Fuhse, 2015a).³

Rather than building on an established theoretical tradition in sociology, relational sociology formulates a specialized theory of social networks. I first present the basic approach and then some recent extensions, including my own. (p. 41)

Social Networks and Meaning

White was central in American structuralist network research of the 1960s and 1970s, developing blockmodel analysis with his students and directly influencing many authors (Scott, 2000, p. 33ff). During the 1980s, White became dissatisfied with a purely structural vision of networks. He wanted to know what a tie, as the basic building block of networks, was, and why and how it meshed with other ties to form larger structures (Mische, 2011, p. 82). White (1992) laid out his answer in the first edition of his book *Identity and Control*: Social networks are not mere patterns of ties. Rather, they have to be thought of, and studied, as constructions of meaning. Networks consist of *identities* that are constructed in *stories*, and thereby related to each other (White, 1992, p. 65ff).

The starting point for these arguments is—as in many social theories—the general uncertainty of social interaction (White, 1992, p. 3ff). Identities strive for “control” and “footing” in social contexts. Social structures, however, emerge only through the observation by others. Control attempts of identities leave a social trace in the form of stories told about them. White’s (1992, p. 6ff) example here is the children’s playground. The collaboration and jousting between children are observed in a web of stories about them and about the relations between them. White thus views social networks as interwoven with *forms of meaning* (identities, stories) and structured by them. Unlike in pragmatism/interactionism, we are not dealing with “subjective meaning” in people’s heads here. Rather, stories about identities and relations are *communicated*, thereby structuring social interaction (White et al., 2007, p. 544).

Over and above identities and ties, White views networks as interwoven with *domains* of cultural forms (Mische & White, 1998, p. 702ff). Domains consist of symbols, language (e.g., specialized argot), norms, etc. These forms of meaning develop in the interaction in networks and differ from one network context to the next. Cultural forms depend on their development and diffusion in networks, just as networks are imprinted by them. White sees network and domain as inextricably intertwined in “netdoms,” and as only analytically separable.

In the two editions of *Identity and Control*, White (1992, 2008) builds elaborate theoretical architecture on these foundations. Rather than discuss complex but rarely used concepts like “disciplines” or “control regimes,” I concentrate here on insights from White and authors around him on the interrelation between network constellations and forms of meaning.

Identities in networks are not prefabricated building blocks, but projection points for their construction in stories. They and their qualities result from interactive control attempts and from their narrative observation. Charles Tilly (2002, pp. 8f, 26f) regards the *attribution of subjective dispositions* to these identities in stories as key for this process. Observed behavior is traced to relatively stable motivations and capacities for action. This leads to expectations about the future behavior of these identities. In this vein, artists are seen as more or less talented (capacity) based on the quality of their paintings (observed behavior) and as adherents of particular styles (motivations). Padgett and Ansell (1993) show the importance of these processes of attribution for the arrangement of social relations in the ascent of the Medici to power.

(p. 42) Network formations bring about particular *styles* and are imprinted by them (White, 2008, p. 112ff). First of all, a style is an artifact of observation: the behavior of multiple identities in a network is classified as similar, and the identities are recognized as adhering to a common style. The classification becomes part of social structure if the control attempts of identities follow the style. For example, artists produce paintings with the characteristics of impressionism, fauvism, or abstract expressionism (White, 1993, p. 63ff). This back and forth between observation, classification, and orientation relates identities to each other as similar or dissimilar. Styles often develop in *dense networks* and demarcate them symbolically. But they can also correlate to *structurally equivalent positions* in networks. For example, the social movement leaders studied by Ann Mische (2008) display particular communication styles according to their affiliation with one or more movement organizations.

Social categories (like gender or ethnicity) group identities by criteria of similarity and difference. White (2008 [1965]) early termed categorically separated networks as “catnets” (for *category* and *network*). Blockmodel analysis reconstructs not only categories dividing dense network clusters from each other but also role categories of structural equivalence. Social categories are connected to rules for interaction within and between them. In White’s theory since 1992, categories become part of the domain of cultural forms in a network. According to Tilly (1998, p. 63f), categories need stories to render them plausible and legitimize them (“boundary stories”). Social categories can emerge as “collective identities” from the cooperative interaction in dense networks and motivate for collective action (Gould, 1995). But they can also be used to bar outsiders from resources (Tilly, 1998, pp. 6f, 75ff).

Paul DiMaggio combines White’s theoretical arguments with those from Pierre Bourdieu (DiMaggio & Powell, 1983; DiMaggio, 1986). *Institutions* (like marriage, shaking hands, or formal organization) are cultural models for interaction. They emerge isomorphically in

the mutual orientation of actors and institutionalize through repetition and imitation. Therefore, they differ by network context. Following DiMaggio (1986), some institutions concern the relations between categories of actors. They can be reconstructed by way of blockmodel analysis. White (2008, p. 171ff) conceptualizes institutions as networks that are strongly patterned by cultural models.

Overall, relational sociology of and around Harrison White does not really provide a strong and consistent perspective, like action theory. Rather, it offers a creative collection of theoretical arguments about the interplay of meaning and networks. Its strength lies in the fruitful connection to empirical social research, as exemplified by the studies of Mische (2008), Padgett and Ansell (1993), Gould (1995), and DiMaggio (1986).

Extensions

Since the end of the 1990s, relational sociology has been extended in two ways:

(1) *Culture* is now itself conceptualized and studied as a network of symbols meaningfully connected to each other (Mohr, 1998; Fuchs, 2001). Some symbols or concepts in a culture (or in the “domain” connected to a network) are more central, others in the periphery. Systematic differences between symbolic universes can be analyzed in (p. 43) their network structures. Generally, relational sociology is most interested in the relations of symbols that are connected to the social networks between actors, for example, in reconstructing the meaning of social categories (Mohr, 1994) and relationship frames (Yeung, 2005), or in identifying “weak culture” shared across network clusters (Schultz & Breiger, 2010).

(2) Relational sociology increasingly turns to the *communicative events* in which networks are negotiated and constructed (Mische, 2003, p. 262; Mützel, 2009). Godart and White (2010) argue that “switchings” between sociocultural contexts make for the emergence of meanings and the change of social structures. Emirbayer (1997, p. 287) and Tilly (2005, p. 6f) refer to “transactions” as the basic process in networks (see Pragmatism and Interactionism), while Padgett (2012) and I (Fuhse, 2015b) build on different notions of “communication.” This second push leads to a concern for studying the dynamics (and the reproduction) of networks in communication, rather than fixed patterns of ties (e.g., by drawing on conversation analysis and/or with relational event models).

Both advancements connect well with the recent turn to big data and computational social science. Relational sociology provides a fruitful backdrop for formal quantitative research on communicative events and on symbolic structures in data compiled from large databases, from journalistic and political discourse, and from the Web 2.0. Authors from the tradition play a prominent role in the theoretical reflection of big data analyses (Mohr, Breiger, & Wagner-Pacifci, 2015).

In my own work, I try to provide a coherent framework for the various threads running through relational sociology (Fuhse, 2009, 2015b, in preparation). Social networks are conceptualized as dual in nature. They consist of (1) *observable regularities in communi-*

cation and (2) the *patterns of meaning* underlying these regularities. This “meaning structure” of a social network is made of “relational expectations” about the behavior of particular actors toward particular others. These arise out of the attribution of communicative events to actors, making for definitions of identities and their relations to each other. In turn, relational expectations structure communication by rendering particular events likely and others unlikely.

Communicative regularities are influenced, of course, by opportunities for contact at foci of activity. In contrast, the meaning structure of networks builds on *cultural models* for relationships and network structures. For example, relationship frames like “love,” “friendship,” or “patronage” make for very different relationships and network structures. Communication and meaning structure thus retain a certain relative autonomy, in spite of their close interplay.

Actors can be not only individual but also formal organizations (e.g., companies, universities, states) and collective actors (e.g., social movements, street gangs). *Collective actors* are constructed on two levels: (1) internally, the sharp drawing of a boundary of meaning to the outside and the symbolic construction of a collective identity go hand in hand with an increased density of ties and the thinning out of personal ties to the outside, and (2) in the network with other collective/corporate actors (e.g., in the political field), they effectively become actors if communicative events (e.g., protest marches for social movements or homicides for gangs) are attributed to them rather than to individual actors. This leads to “relational expectations” about how collectives behave toward other collective or corporate actors.

(p. 44) These various advancements lead to a complex theoretical architecture with social networks as patterns of communicative events and intertwined with forms of meaning (that can themselves be studied in networks of symbols). Overall, relational sociology is mostly connected to the formal study of full networks, with blockmodel analysis and recently exponential random graph models as methods of choice (Gondal & McLean, 2013). Cultural networks are sometimes studied with topic modeling or Galois lattices. But, like the pragmatist and interactionist approaches, relational sociology also connects to qualitative methods (Fuhse & Mützel, 2011).

Conclusion

The approaches termed here “theories of social networks” are necessarily selective, and their presentation remains sketchy and stylized. Due to space constraints, I had to leave out approaches like network exchange theory (Emerson, Cook) and constructural theory (Carley, Mark). Table 3.2 gives an overview of the three families of action theory, pragmatism/interactionism, and relational sociology.

Table 3.2 Theories of Social Networks

	Theory of Action/Rational Choice	Pragmatism/Interactionism	Relational Sociology
Key authors	Burt, Coleman, Lin, Hedström	Emirbayer, Crossley, Martin	White, Tilly, Mische, Padgett, Fuhse
What are social networks?	Opportunities, constraints	Interaction patterns, definition of situation	Identities and stories, relational expectations
Locus	Objective structure/individual resource	Subjective meaning/interaction	Ties, transactions/communication
Methods	Ego-centric networks, SIENA, agent-based modeling	Qualitative interviews, relational ethnography	Full networks, blockmodel analysis, Galois lattices

Action theory views social networks in a naturalist way as objective structures that afford individuals with opportunities and constraints, depending on their position and embeddedness. The *social capital* concept reduces these opportunities and constraints to a resource. Individual actors can have more or less of that resource. With their individualist perspective, action theory and the social capital approaches lend themselves to combination with the statistical analysis of ego-centric networks, as well as with agent-based modeling and with SIENA.

The authors adhering to *pragmatist and interactionist perspectives* (Fine and Kleinman, Emirbayer, Crossley, Martin) get away from networks as objective structures. They locate them in the minds of actors and in the interaction between them. Social networks consist in the subjective/intersubjective patterns of meaning attached to ties (definitions of the situation). With their focus on meaning, these approaches connect well with the qualitative study of networks in interviews and in “relational ethnography.”

(p. 45) In *relational sociology* by Harrison White and his colleagues and students, networks consist of identities that are related to each other in stories. In spite of its strong connections to pragmatist and interactionist authors, relational sociology avoids the minds of individuals. Instead, it regards social networks as supra-individual structures of meaning, or as patterns of transactions/communication. Relational sociology studies

mostly full networks with formal methods like blockmodel analysis. With the extensions to cultural networks and to communicative events, relational sociology also draws on Galois lattices, topic modeling, and sequential and qualitative analyses of communication.

To conclude this overview, I list a number of challenges for theories of social networks:

(1) All the approaches covered here move *away from purely structural or formal notions of networks* and relations (Erikson, 2013). In different ways, the three theoretical families conceptualize social relations as *processual* and *dynamic*. The basic processes at play are termed action, interaction, transaction, or communication, with relations and networks as historical snapshots, results, and determinants. The concepts of action and agency make individuals the sources of these processes, while transactions and communication are distinctly supra-personal. Interaction occupies a middle ground between those two poles. Interactionist and pragmatist approaches point to the importance of subjective *meaning* (as well as of its interactive negotiation), while relational sociology focuses on meaning that is communicated/translated. Action theory at times incorporates a similar concern for subjective meaning. But the concept of social capital is devoid of these considerations.

(2) Who, or what, are the actors in networks? Action theory, pragmatism, and interactionism build on individuals as processing meaning/deciding about their actions. Consequently, they theorize social networks primarily or solely between individuals. Some authors propose their frameworks to be compatible with higher-level actors (like companies or social movements), but without accounting for them theoretically (e.g., Crossley, 2010a, p. 43f). For relational sociologists, networks are bundles of stories/expectations that build on the attribution of events to actors, be they individual, collective, or corporate. This leads to a multilevel architecture of the social world, with social relations located on various levels, and with important repercussions between them (Fuchs, 2001; White, 2008).

Material objects or *cultural symbols* do not feature as actors in any of the theories of social networks surveyed here. They do not take decisions (action theory) or process meaning subjectively (pragmatism/interactionism). Nor are stories told about them or relational expectations attached to them (relational sociology). To address cultural relations between symbols, or sociotechnical relations between objects and human actors, we need different theories.

(3) The theories covered here provide highly general frameworks. However, they do not give much indication what network configuration we will find empirically or how they will evolve over time. Over the last years, this desideratum has been addressed under the rubric of “network mechanisms” like reciprocity, homophily, foci of activity, transitivity, preferential attachment, brokerage, and contagion. Only a few of the theoretical approaches link to network mechanisms. According to Hedström (2005), all social mechanisms should be modeled on the level of individual decisions. Tilly (2005, p. 23ff) disagrees: mechanisms are chains of transactions that recurrently lead (p. 46) to a change in social configurations—without invoking inner-directed decisions. Martin (2011) argues that actors adopt simple heuristics of interaction de-

pending on circumstances and on their positions in fields. These heuristics correspond to network mechanisms.

(4) The attempts to connect to Bourdieu's field theory point to the need for combining network research with concepts for *macro-structures of society* like politics, art, the economy, science, etc. Action theory remains notoriously silent about these macro-phenomena. Pragmatists and interactionists like Crossley, Emirbayer, Desmond, and Martin, and neo-institutionalist DiMaggio pick up on the concept of fields. Relational sociology seems to provide for this macro-connection with its multi-level architecture of networks. This begs the question of how the interrelation between different levels should be modeled: how do processes within and between social units affect each other?

(5) Finally, we have to consider what we understand by "theory" and what we expect from theorizing about social networks (Abend, 2008). Broadly speaking, scientific theories are compounds of meaningfully connected sentences that are conceptually and logically coherent, and that include statements about empirically observable phenomena. They should be highly general and sufficiently abstract to cover a wide range of phenomena (as opposed to theories of the middle range). However, we are looking for theories of *social networks* as a concept developed out of empirical studies. Therefore, theories should not only be coherent and abstract but also connect to empirical research, to provide theoretical expectations and a framework for the interpretation of results. As pointed out, the theories under consideration differently connect to research methods: action theory and the concept(s) of social capital inform statistical analyses of ego-centric networks. Interactionist and pragmatist approaches lean toward participant observation and qualitative interviews. And relational sociology is most often combined with formal network analysis.

At the same time, the expectations in theorizing differ profoundly, and they sometimes seem to drive the choice for one theory or another. Action theory and pragmatist/interactionist approaches build on a commitment to individual actors. Often, this commitment seems to stem from ontological beliefs and/or from normative impetus. For example, action theory insists that only human beings have "causal powers" (Coleman, 1990, p. 4; Hedström, 2005, pp. 28, 34). In a similar vein, Emirbayer and others emphasize agency as the "moment of freedom" of human actors (Emirbayer & Goodwin, 1994, p. 1142). Relational sociology, in contrast, deliberately strays from everyday thinking and language into its very own realm of concepts, arriving at highly abstract assertions about the social world with individuals as secondary to supra-personal forces. My sympathies lie with this approach, since ontological assumptions and normative considerations do not necessarily lead to sound theoretical statements—even if the epistemic break with everyday conceptions comes at the expense of intuitive plausibility.

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Notes:

- (1.) Charles Kadushin (2002) derives a similar distinction between individual motivations conducive to dense network structures or to bridges across structural holes from the psychological literature.
- (2.) In his later book with Desmond, Emirbayer replaces “transactions” with “interactions” as one component of social configurations (rather than their basic units; Emirbayer & Desmond, 2015, p. 188ff). Also, the authors equate dynamics and process with agency, rather than with supra-personal “transactions.”
- (3.) “Relational sociology” is used here in a much narrower sense than in Emirbayer’s (1997) “Manifesto for a Relational Sociology.”

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